HDOC DAY – 49

Algorithm: Pivot Integer

1. Start.
2. Declare integer variables n, x, i, leftSum, and rightSum.
3. Read the positive integer n from the user.
4. Pass the address of n to the function pivot_integer() using &n.
5. In the pivot_integer() function, receive the address of n using pointer *n.
6. Repeat the following steps for x from 1 to n:
    - Set leftSum = 0.
    - Set rightSum = 0.
    - Using a loop, calculate the sum of all integers from 1 to x and store it in leftSum.
    - Using another loop, calculate the sum of all integers from x to n and store it in rightSum.
    - Compare leftSum and rightSum.
    - If leftSum == rightSum, display "Pivot integer: x" and return from the function.
7. If the loop finishes without finding a pivot integer, display -1.
8. Stop.

C PROGRAM:

```
#include <stdio.h>


void pivot_integer(int *n)
{
   int x, i;
   int leftSum, rightSum;


   for (x = 1; x <= *n; x++)
```

```
  {
    leftSum = 0;
    rightSum = 0;

    // Sum from 1 to x
    for (i = 1; i <= x; i++)
    {
      leftSum += i;
    }

    // Sum from x to n
    for (i = x; i <= *n; i++)
    {
      rightSum += i;
    }

    if (leftSum == rightSum)
    {
      printf("Pivot integer: %d\n", x);
      return;
    }
  }

  printf("-1\n");
}

int main()
{
```

```
    int n;

    printf("Enter a positive integer n: ");
    scanf("%d", &n);

    if (n <= 0)
    {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    // Passing the address of n
    pivot_integer(&n);

    return 0;
}
```

Test Case Table

Test Case	Input n	Expected Output	Reason
1	8	Pivot integer: 6	1+2+3+4+5+6 = 6+7+8 = 21
2	1	Pivot integer: 1	Both sums are 1
3	4	-1	No pivot integer exists
4	49	Pivot integer: 35	Sum 1…35 = Sum 35…49
5	2	-1	No pivot integer exists